

SYDNEY SUN
PORTFOLIO OF WORKS
2021 - 2025

CARNEGIE MELLON UNIVERSITY
BACHELOR OF ARCHITECTURE



Library and Cultural Center 1-6

Mechanic Coffee Patio 7-8

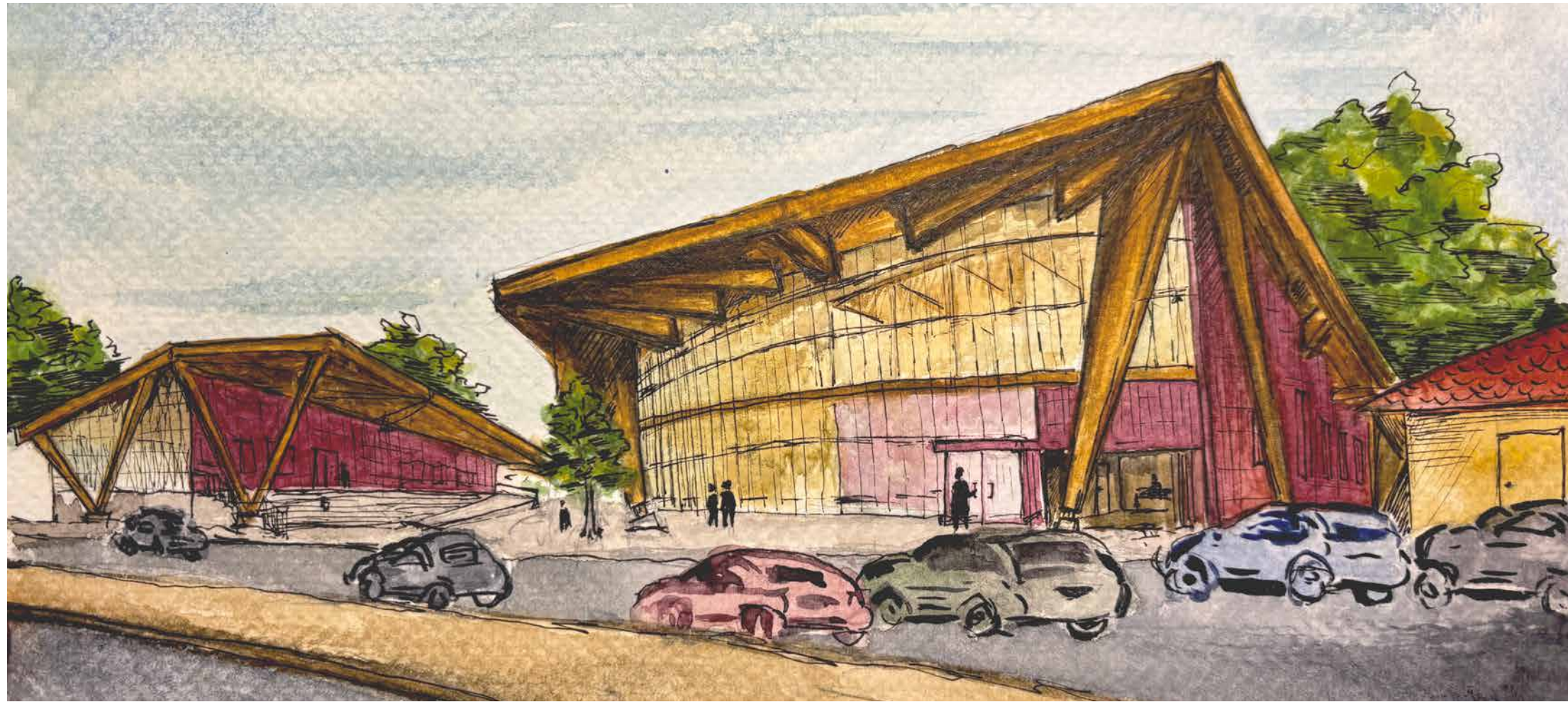
Domesticating Bigness 9-12

The Nodule - Cavern Museum 13-16

Project Slow-Grow 17-22

Fashion Design 23-24

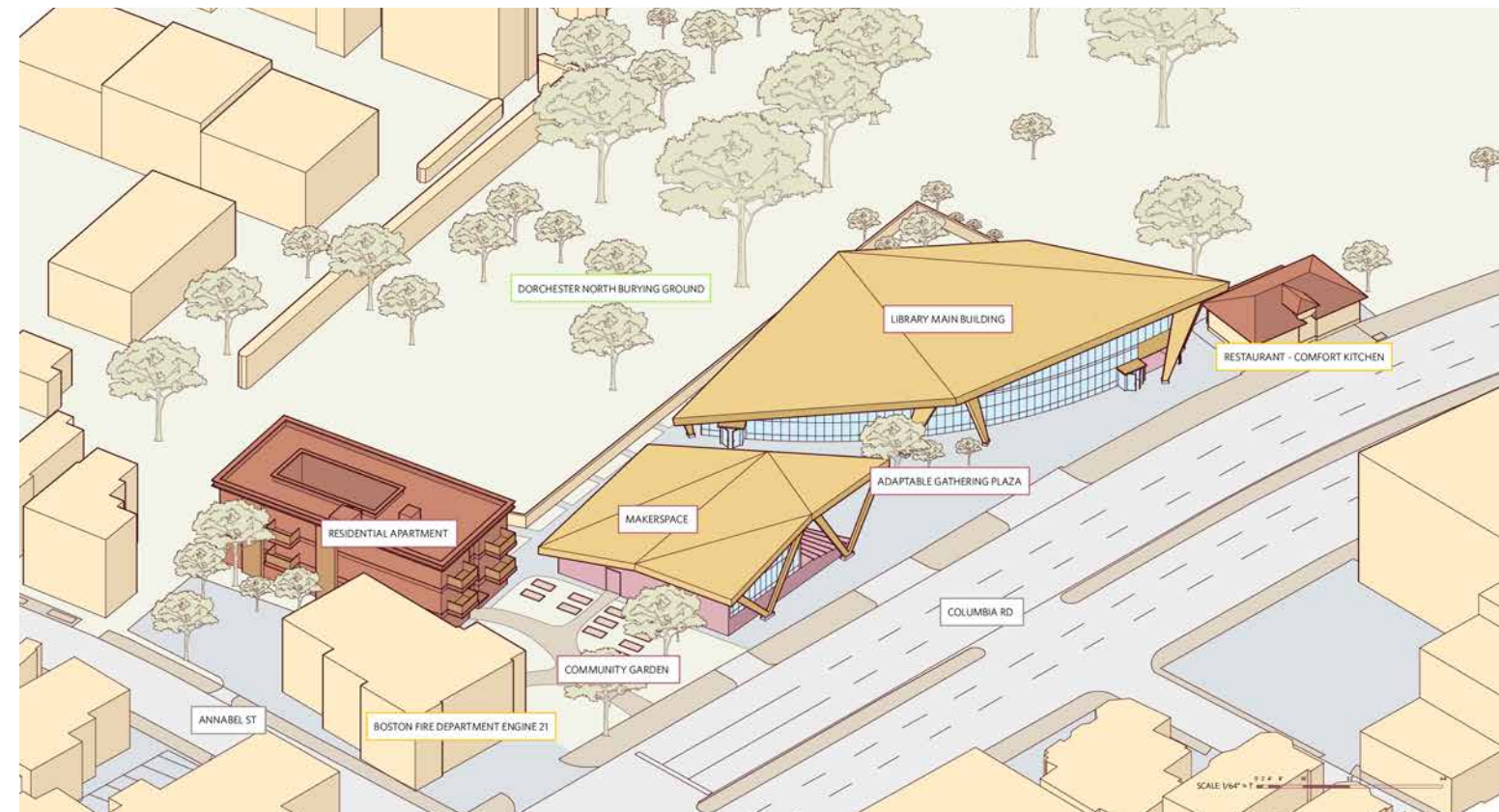
Booth Building 25-26



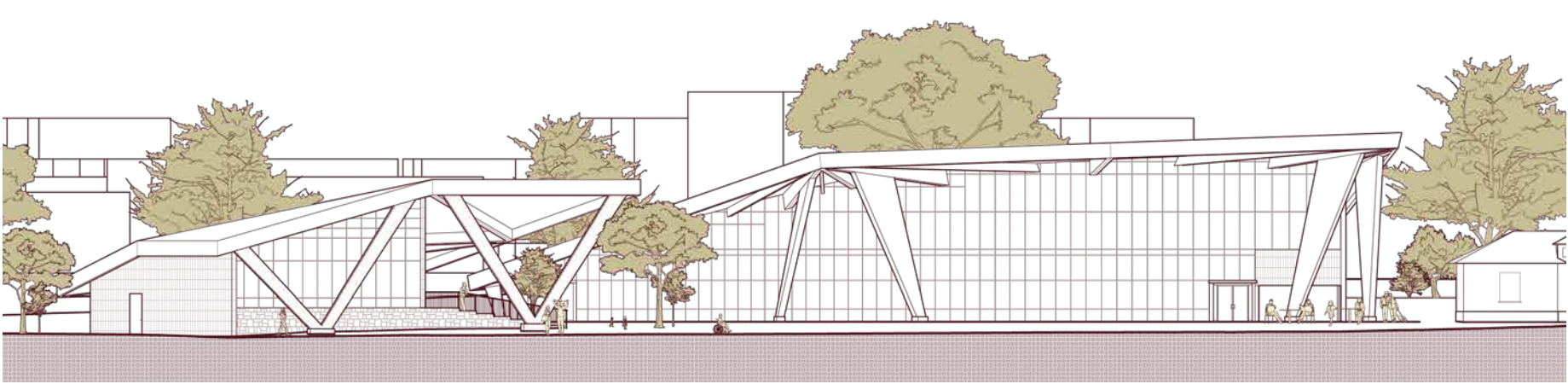
Uphams Corner Library and Cultural Center redefines and emphasizes the value of a library in modern times. The cultural center fosters a welcoming, close knit community in Boston's historic Dorchester neighborhood -- a culturally diverse and artistic neighborhood. The cultural center is divided into four main areas: the main plaza, a library, a makerspace, and an apartment building situated at the back of the site.

The street facing, west-facing facades are all glass double skin facade systems. Not only does this allow for exposure and connection with the public, the double skin facade system operates to passively regulate temperatures in the building, reducing the mechanical heating and cooling load. Coupled with the mass timber construction of the building structure, this building aims to reduce the negative environmental impacts of architecture.

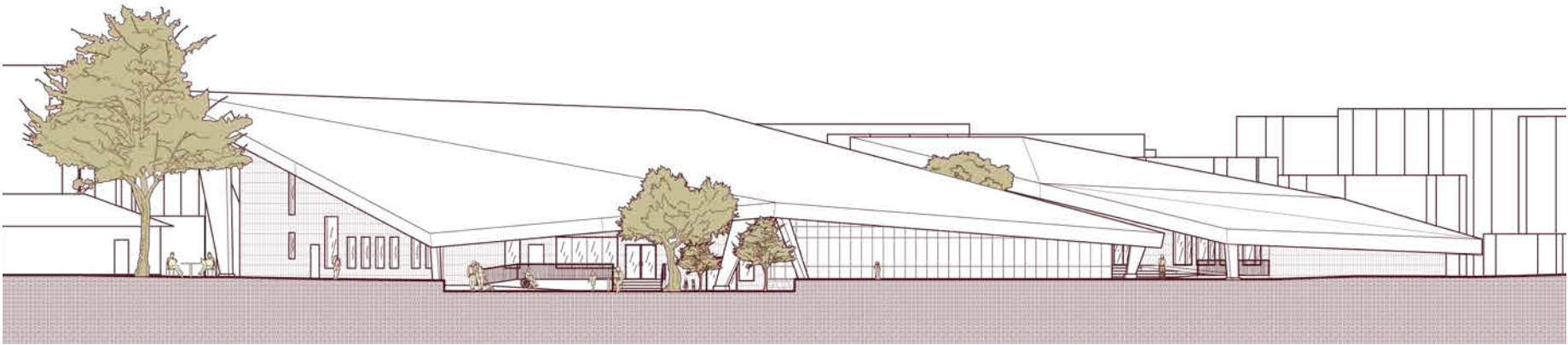
The circulation of the building emphasizes the connection to the public, while allowing for more secluded moments further within the structure. Though the front facade encourages connection, the east side of the library faces the nearby cemetery park, a more tranquil and private area that provides solitude.



SITE DIAGRAM

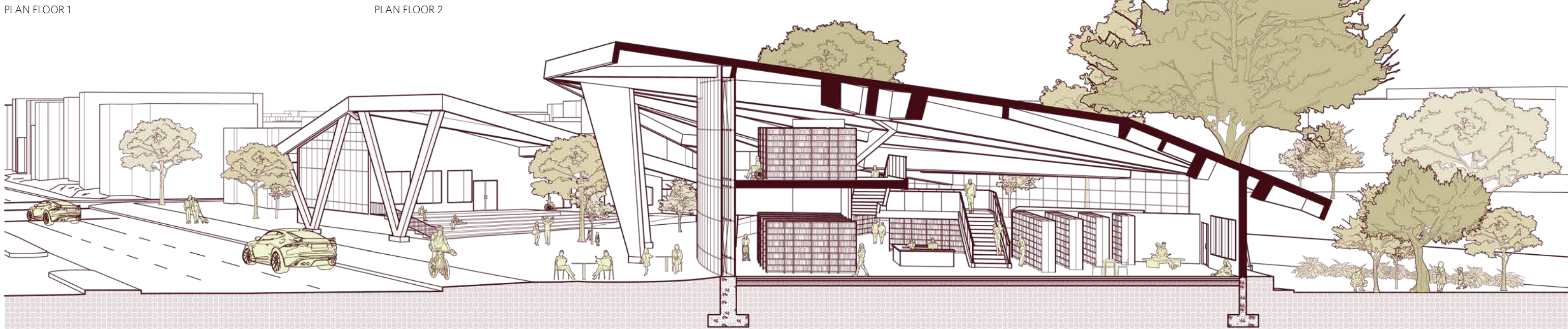


ELEVATION B-B



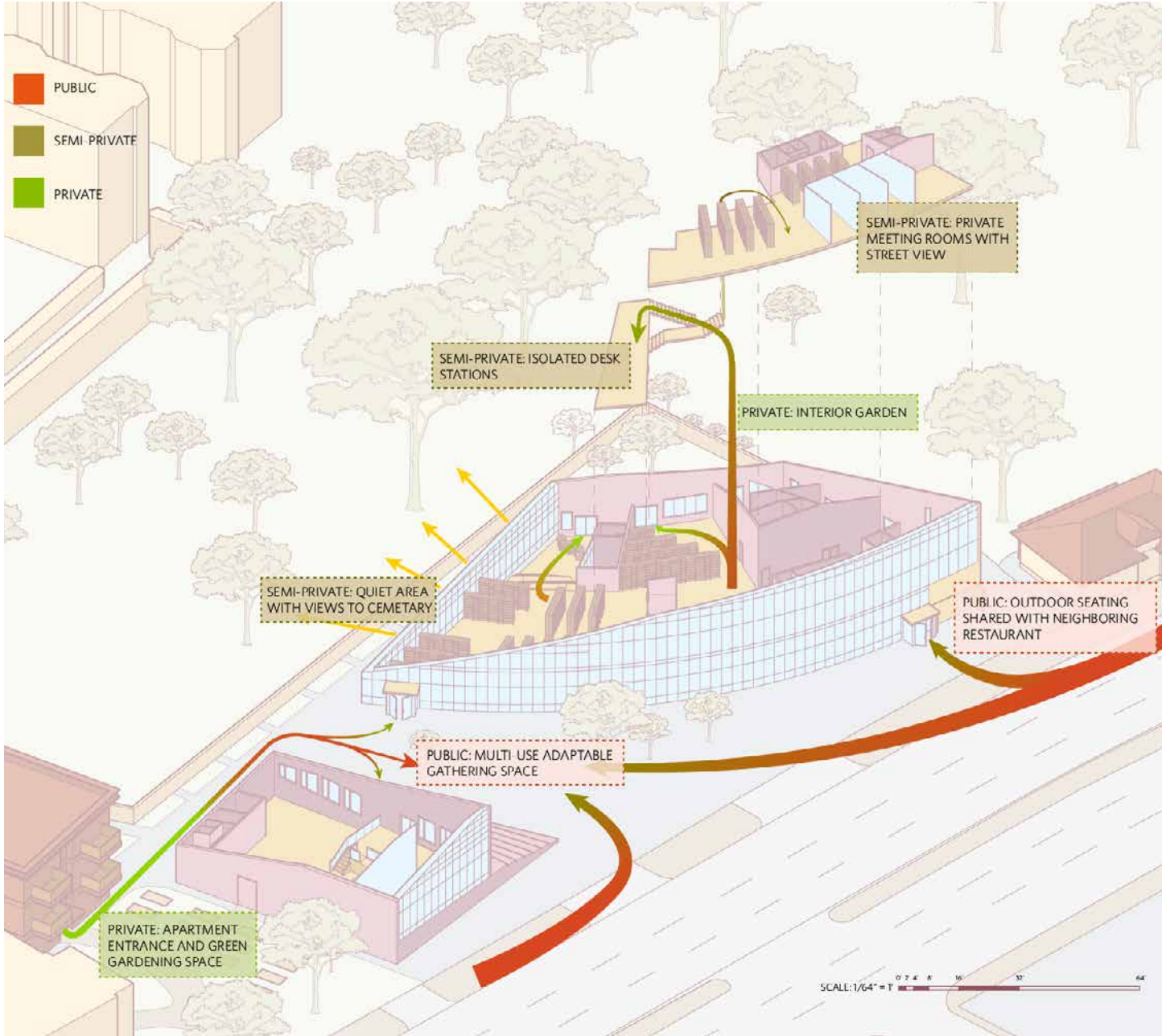
ELEVATION C-C

SCALE: 1/32" = 1' 0' 2' 4' 8' 16' 32' 64'

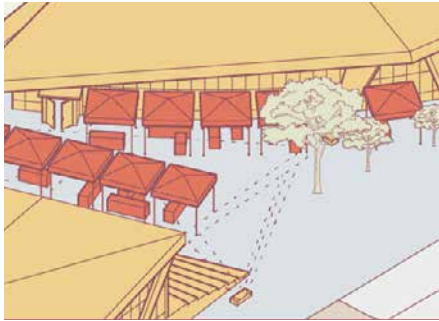


SECTION A-A

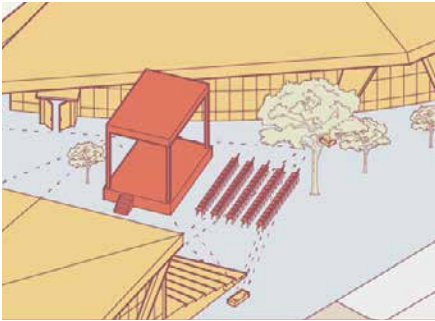
SCALE: 1/16" = 1' 0' 2' 4' 8' 16' 32' 64'



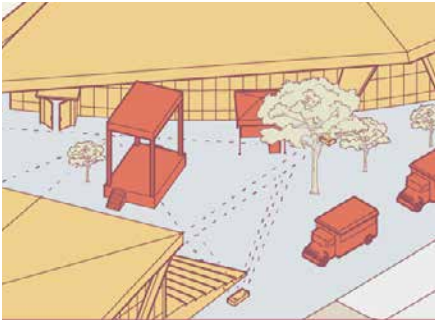
PUBLIC-PRIVATE RELATIONSHIP



- FARMERS MARKET
- ARTIST MARKET
- FLEA MARKET

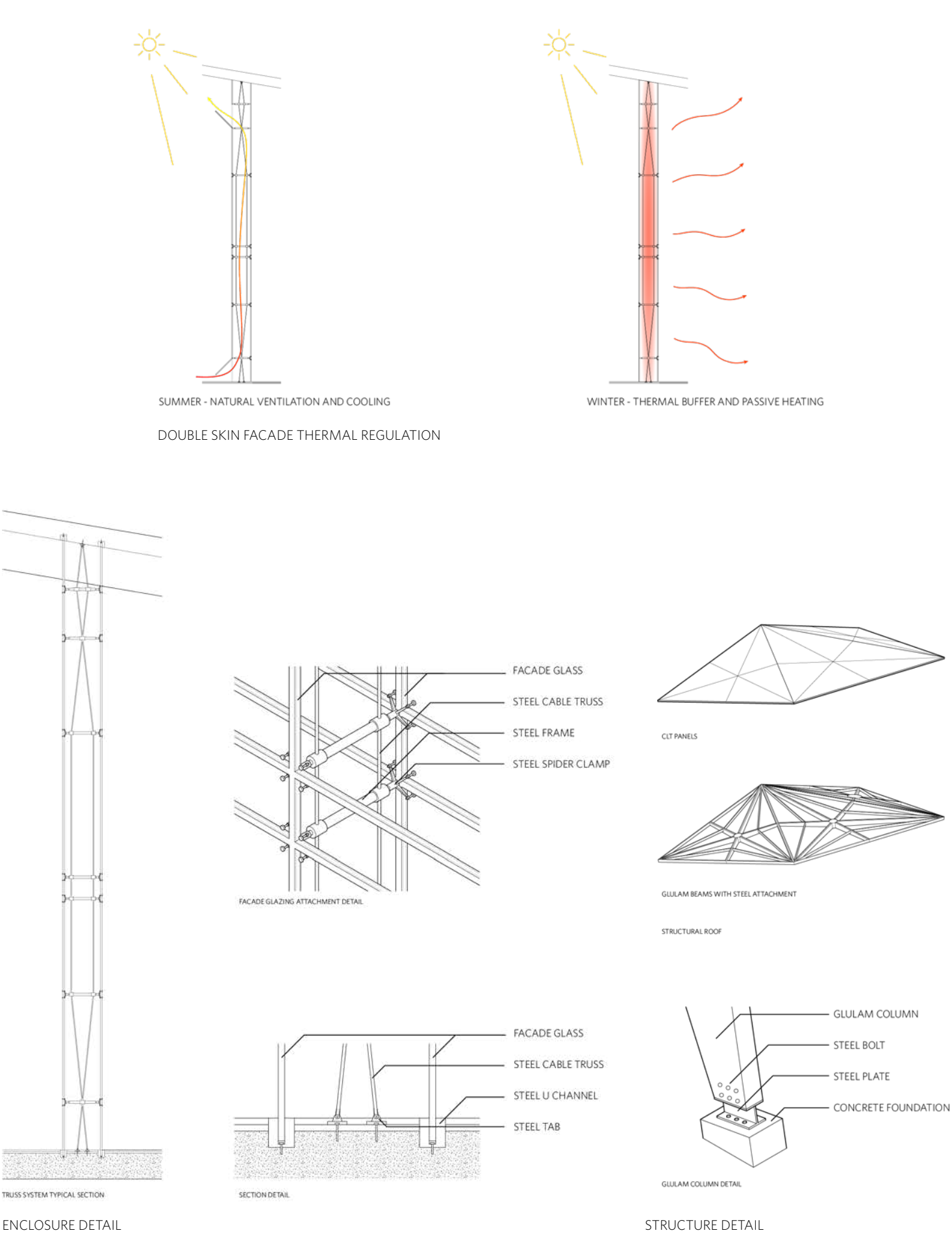


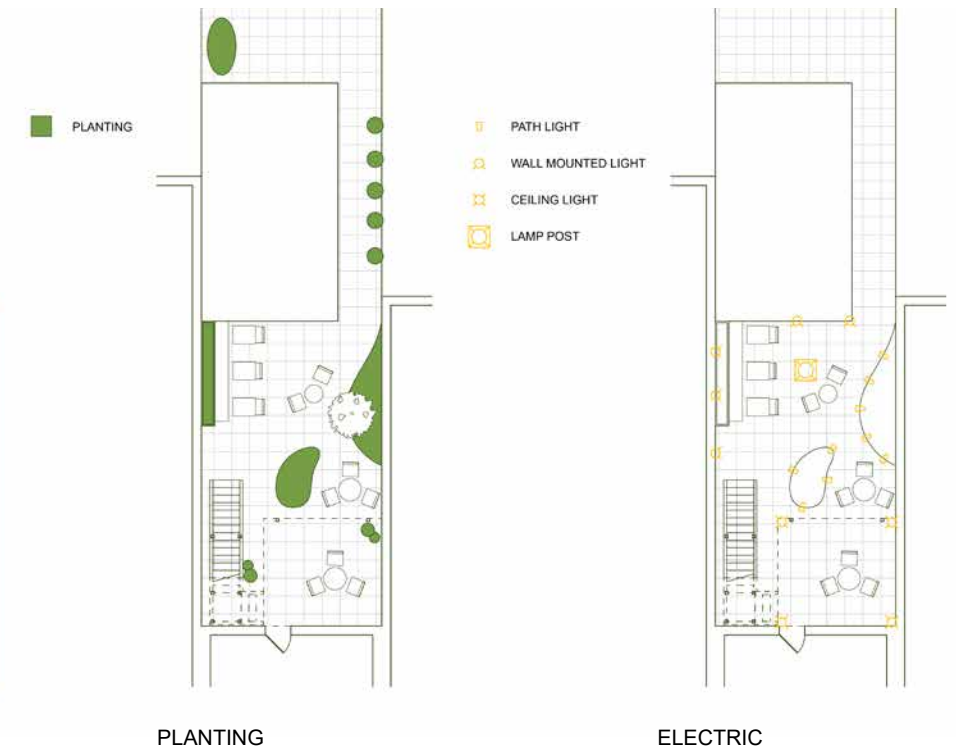
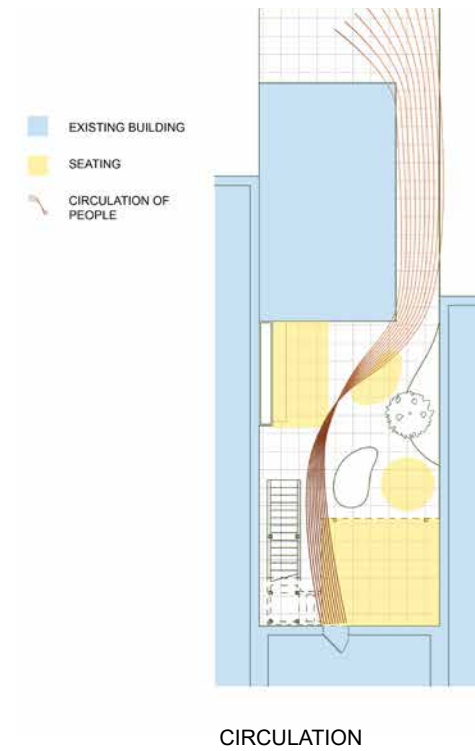
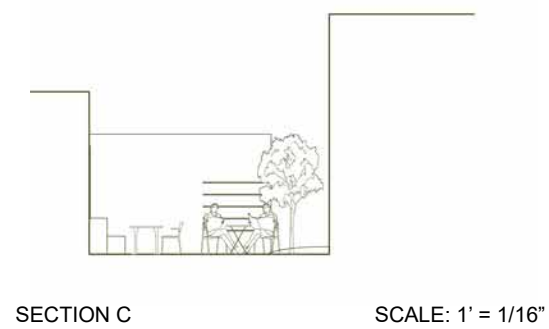
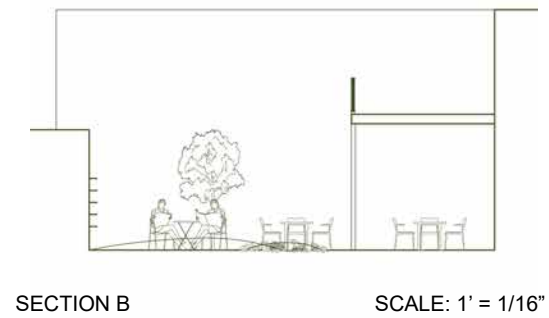
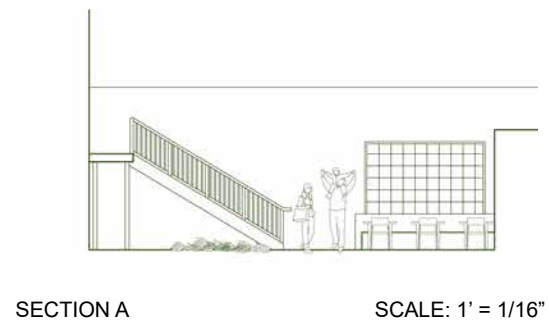
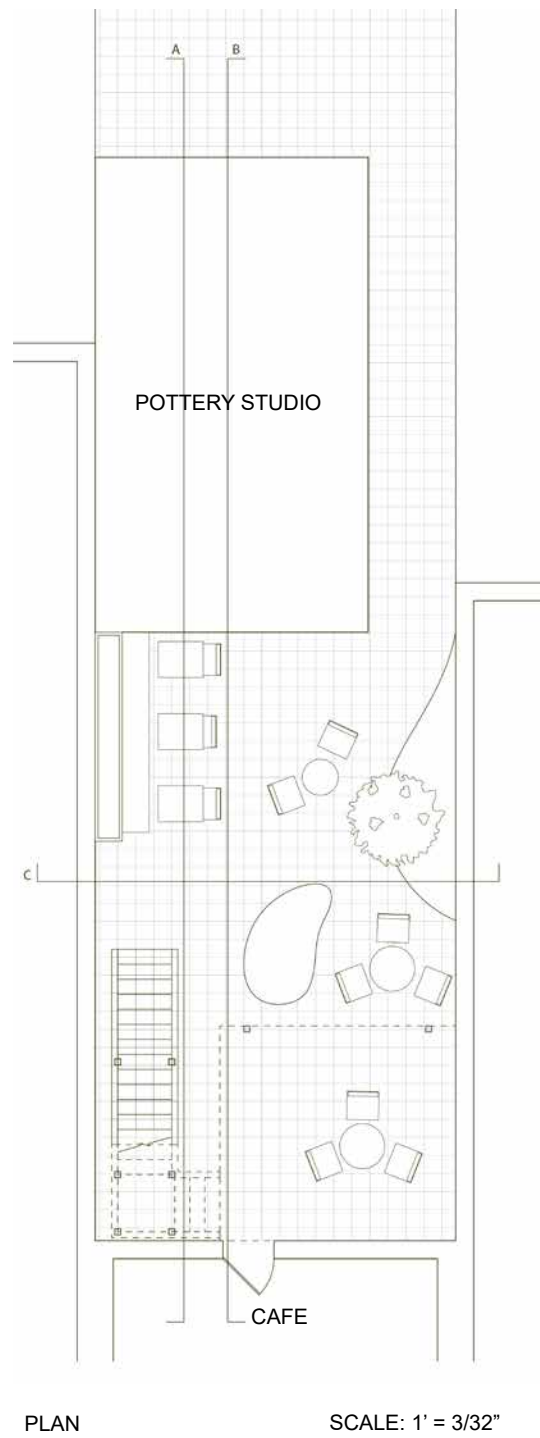
- MUSICAL CONCERT
- CULTURAL PERFORMANCE
- SPEAKING EVENT



- BLOCK PARTY
- HOLIDAY FESTIVAL

GATHERING PLAZA POTENTIAL LAYOUT AND EVENTS





Mechanic Coffee Patio is a patio addition between the main cafe building and the pottery studio behind the cafe. Irregularly shaped planting areas create pockets of seating, inviting customers to immerse themselves in the garden. A single mid-sized tree provides an anchor for the space, while simultaneously shielding view of the driveway beyond the path.

Construction is underway for the patio and is projected to be finished by the end of November 2025.



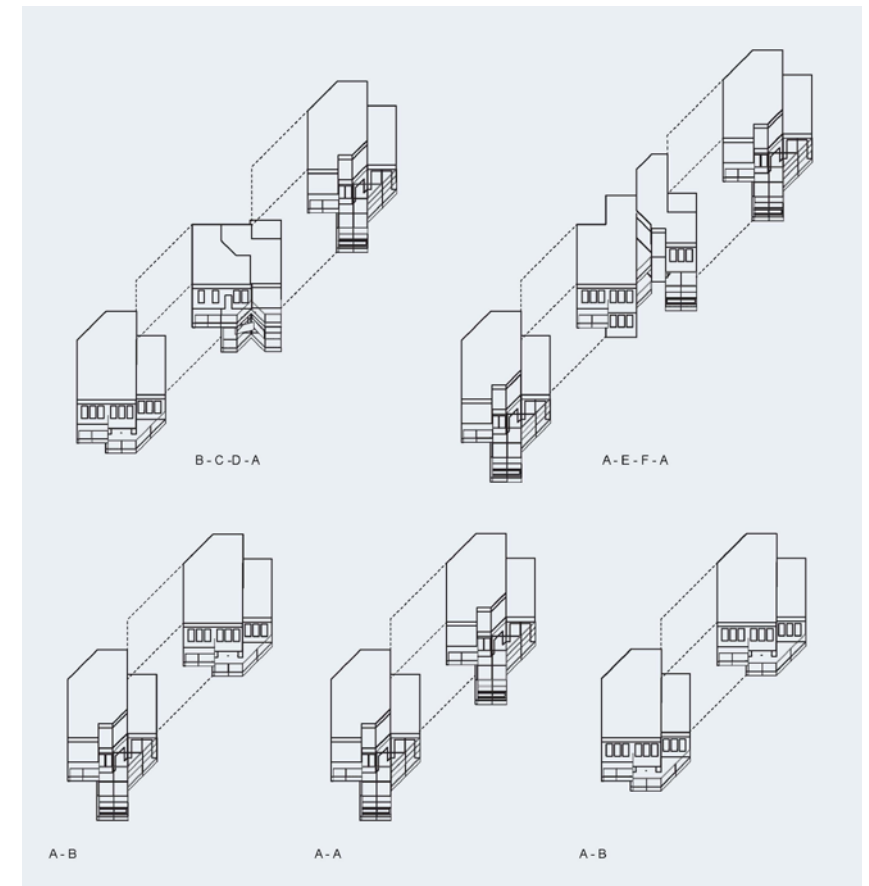


Domesticating Bigness is a modular social housing project that revolves around three levels of urban agriculture -- micro scale of individual private gardens, mezzo scale of communal contribution, and macro scale that involves and welcomes the general public.

The housing community has two different typologies -- courtyard and tower. Each typology is made up of six different housing units, made for residents with different needs, as well as three different communal units. The units are designed on a 20' x 20' structural grid, made for interlocking and aggregation.

The design process began through a flow-chart, categorizing core design concepts, and how the elements are connected. The flow-chart evolves into a program for the project.

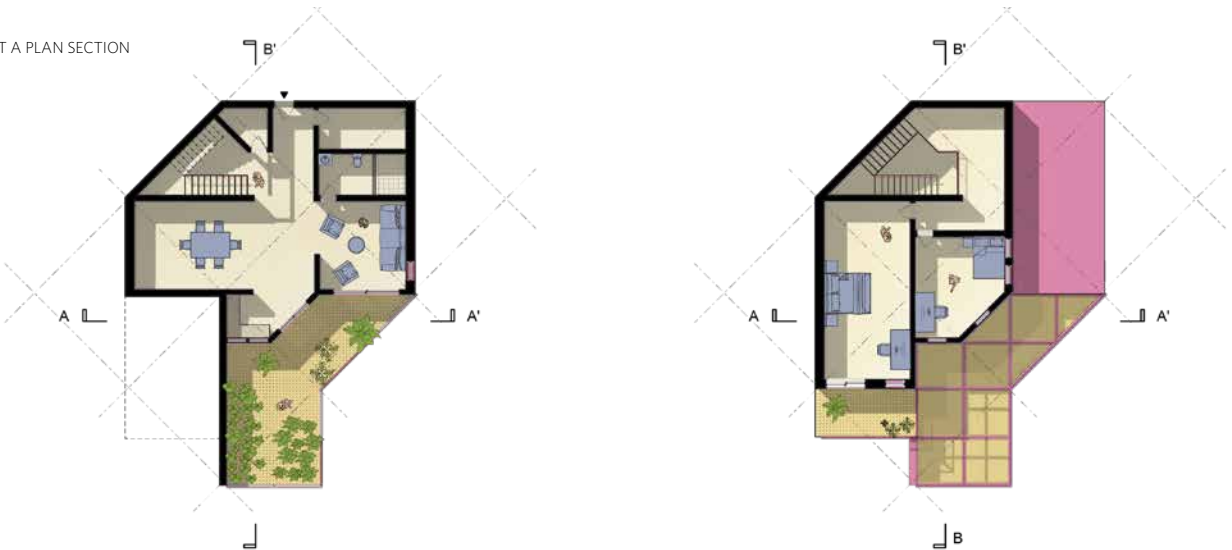
AGGREGATION DIAGRAM



UNIT PLAN SECTIONS



UNIT A PLAN SECTION



SECTION A-A'



SECTION B-B'



COMMUNAL UNIT PLAN SECTION



COURTYARD TYPOLOGY

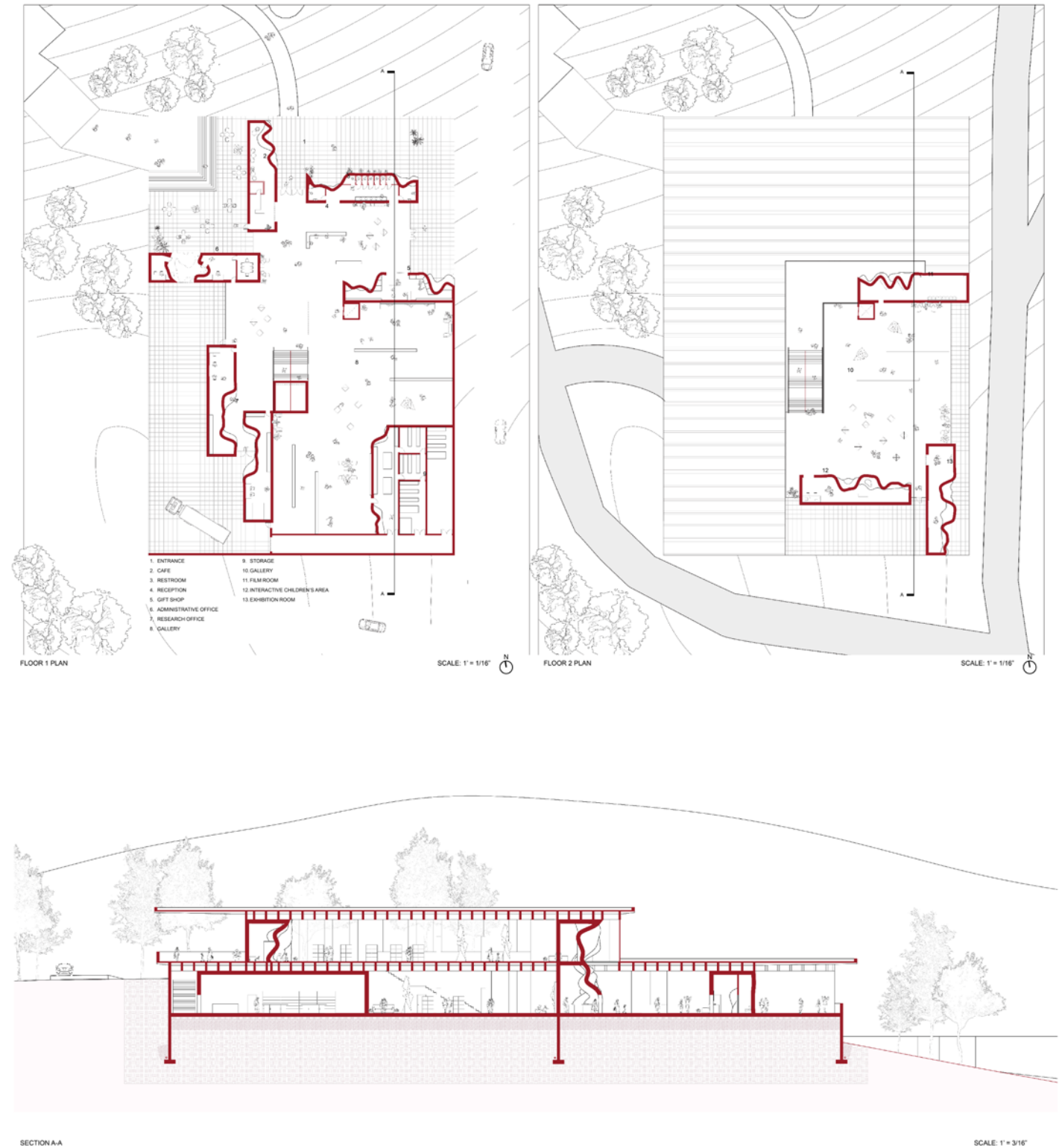


TOWER TYPOLOGY



INTERIOR RENDERS





METHOD 1

Cast in parts to examine Sectional Quality
Use "Crumbled Paper Balls" as infill to simulate stone texture

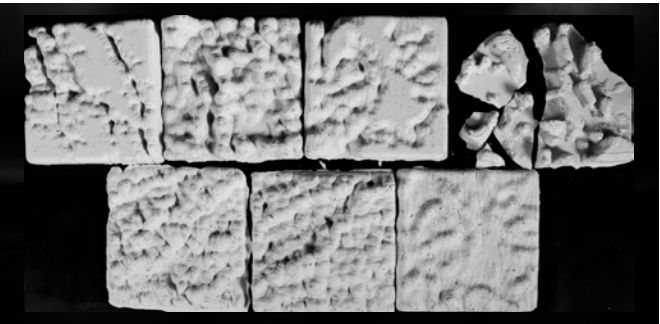
Casting Results



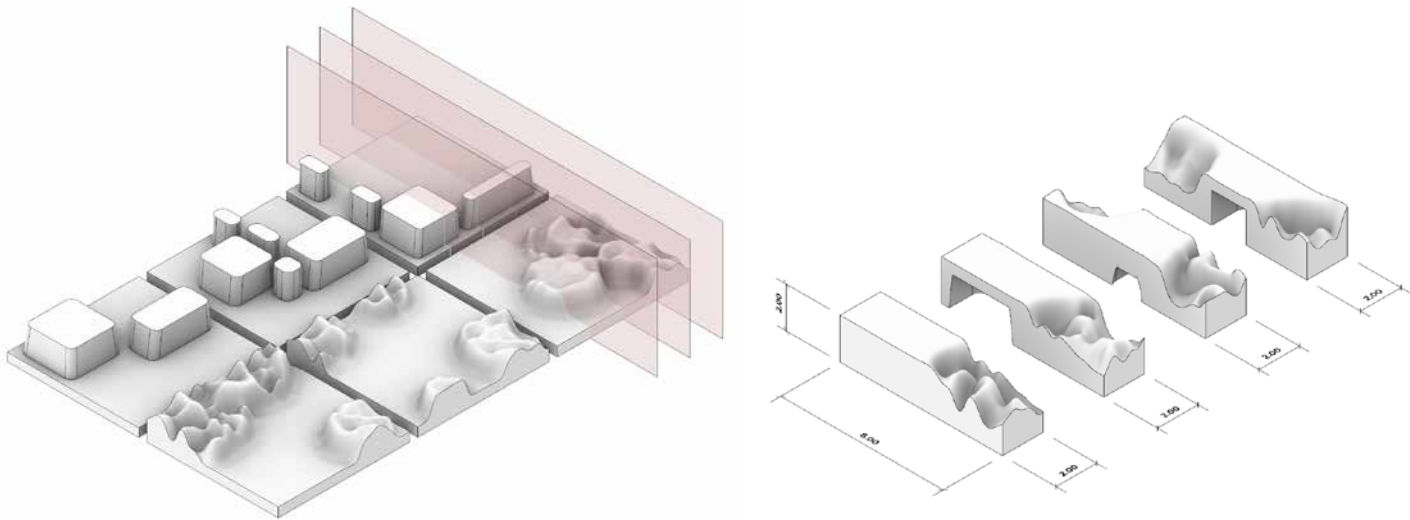
METHOD 2

Using texture maps for project different heights

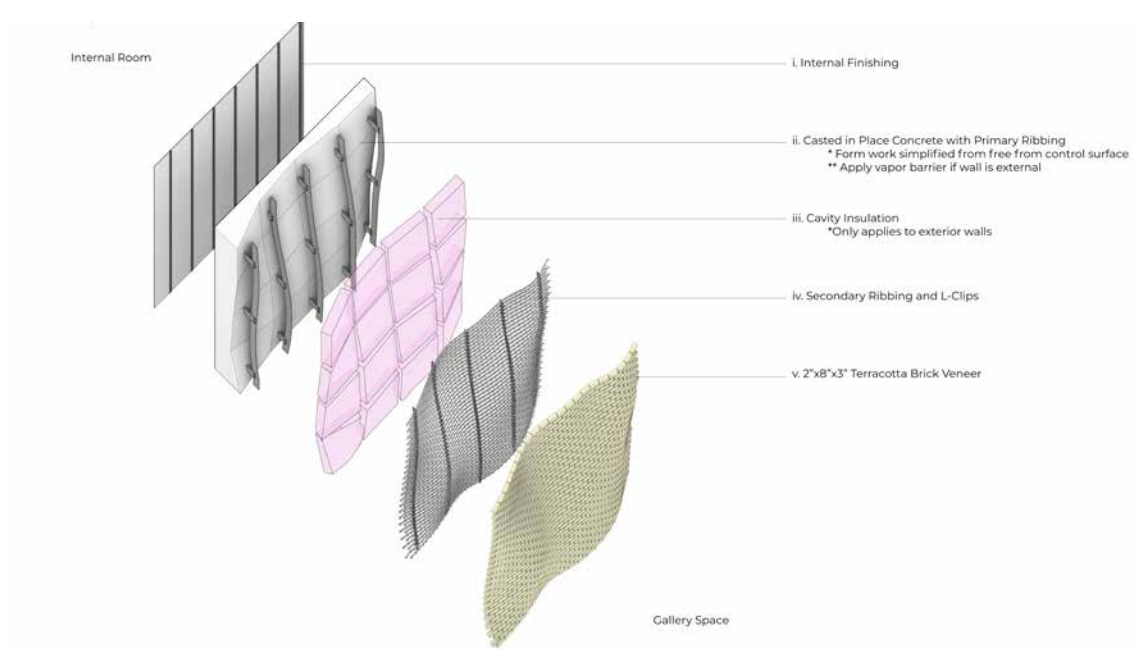
Casting Results



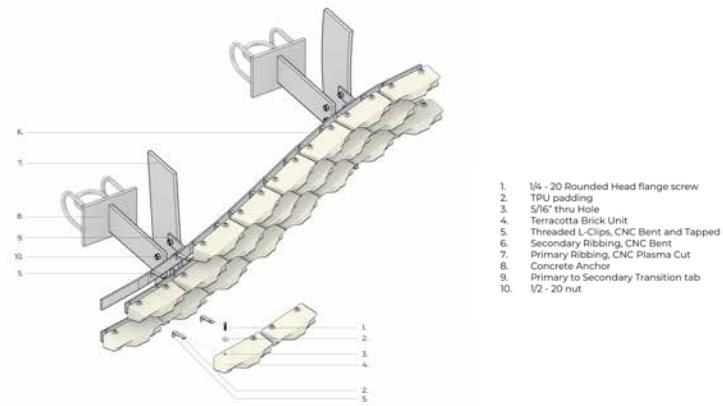
FINAL CASTING FORM BASED ON COMBINATION OF METHODS



COMPOSITION



ASSEMBLY DETAIL



ASSEMBLY MODEL



FINAL CAST



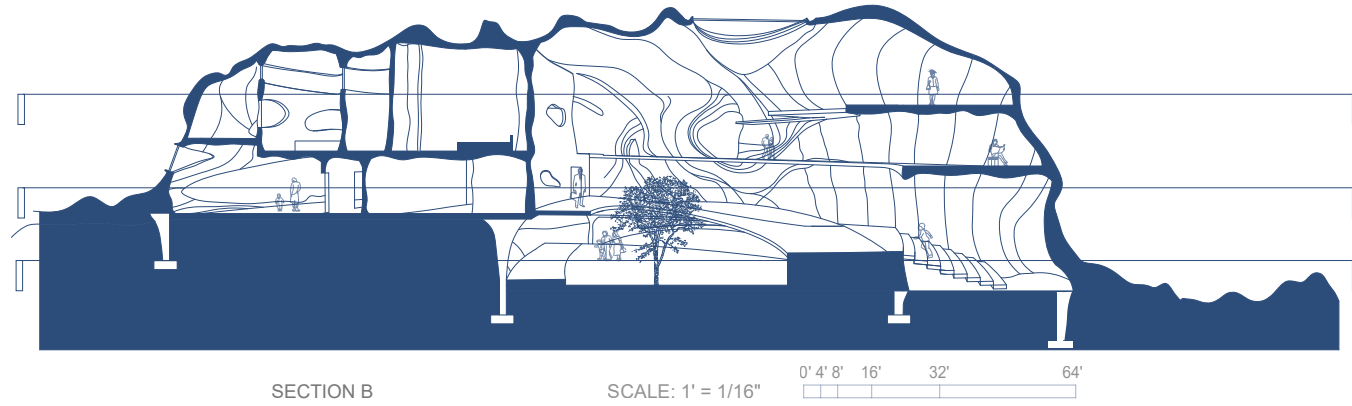
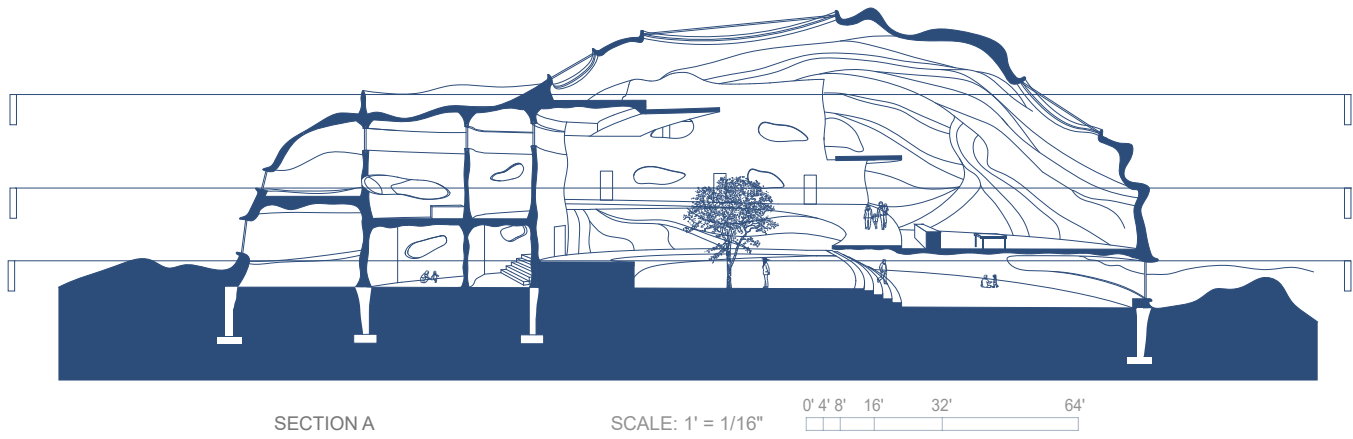
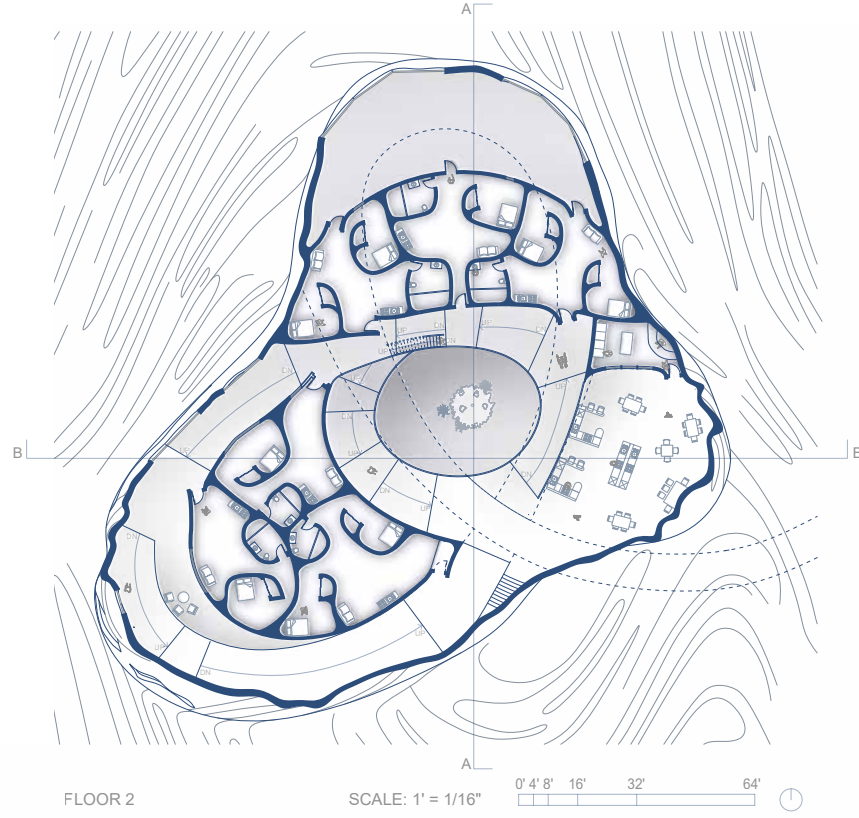
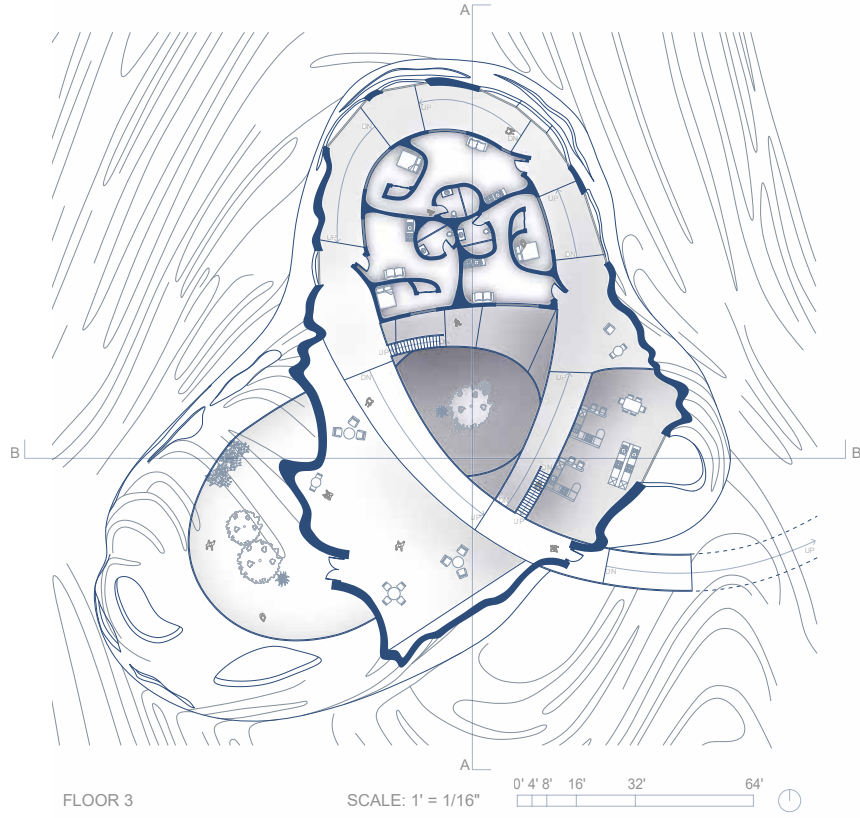
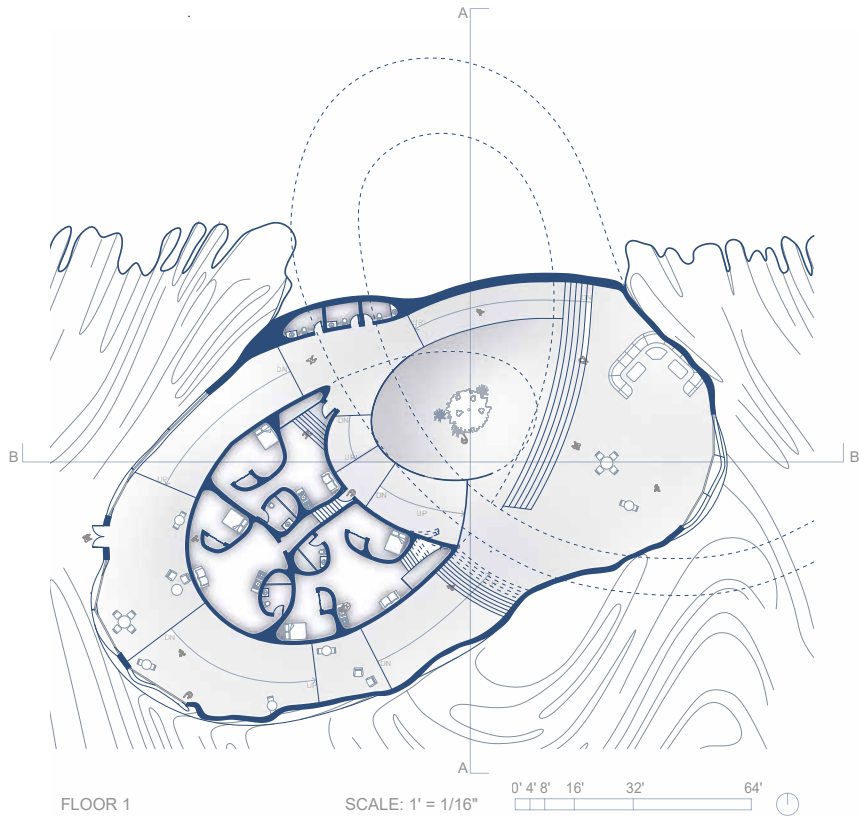
PREVIOUS MOLD ITERATIONS

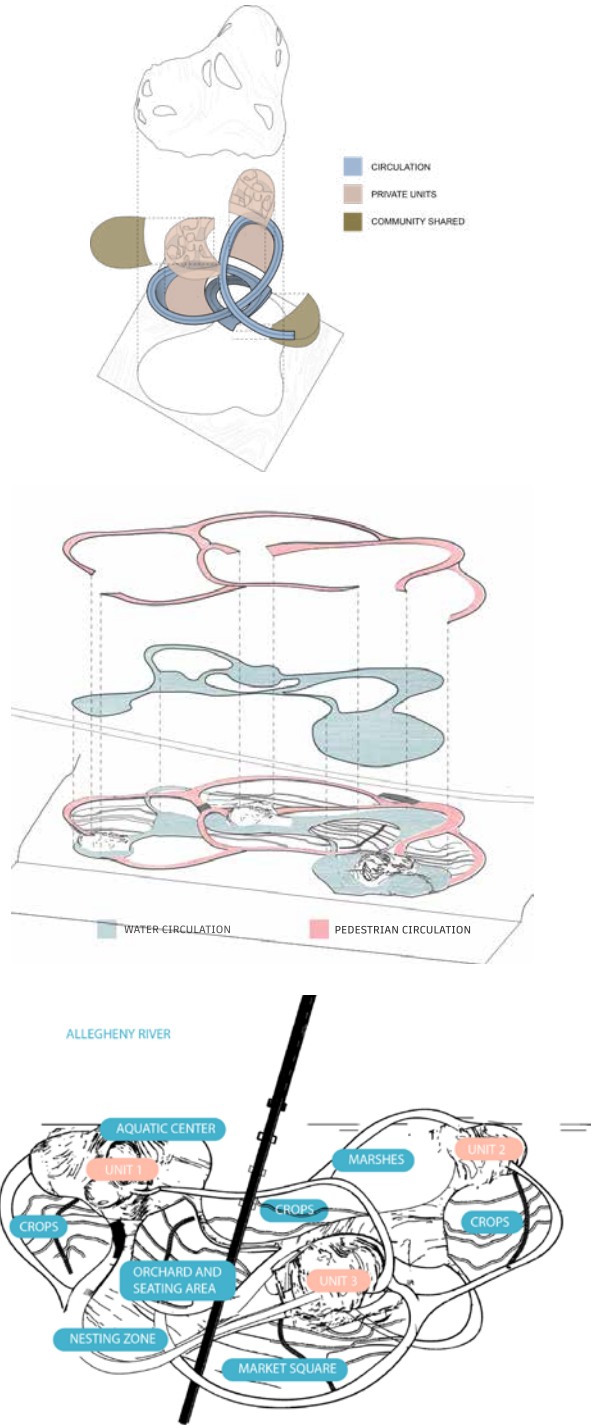
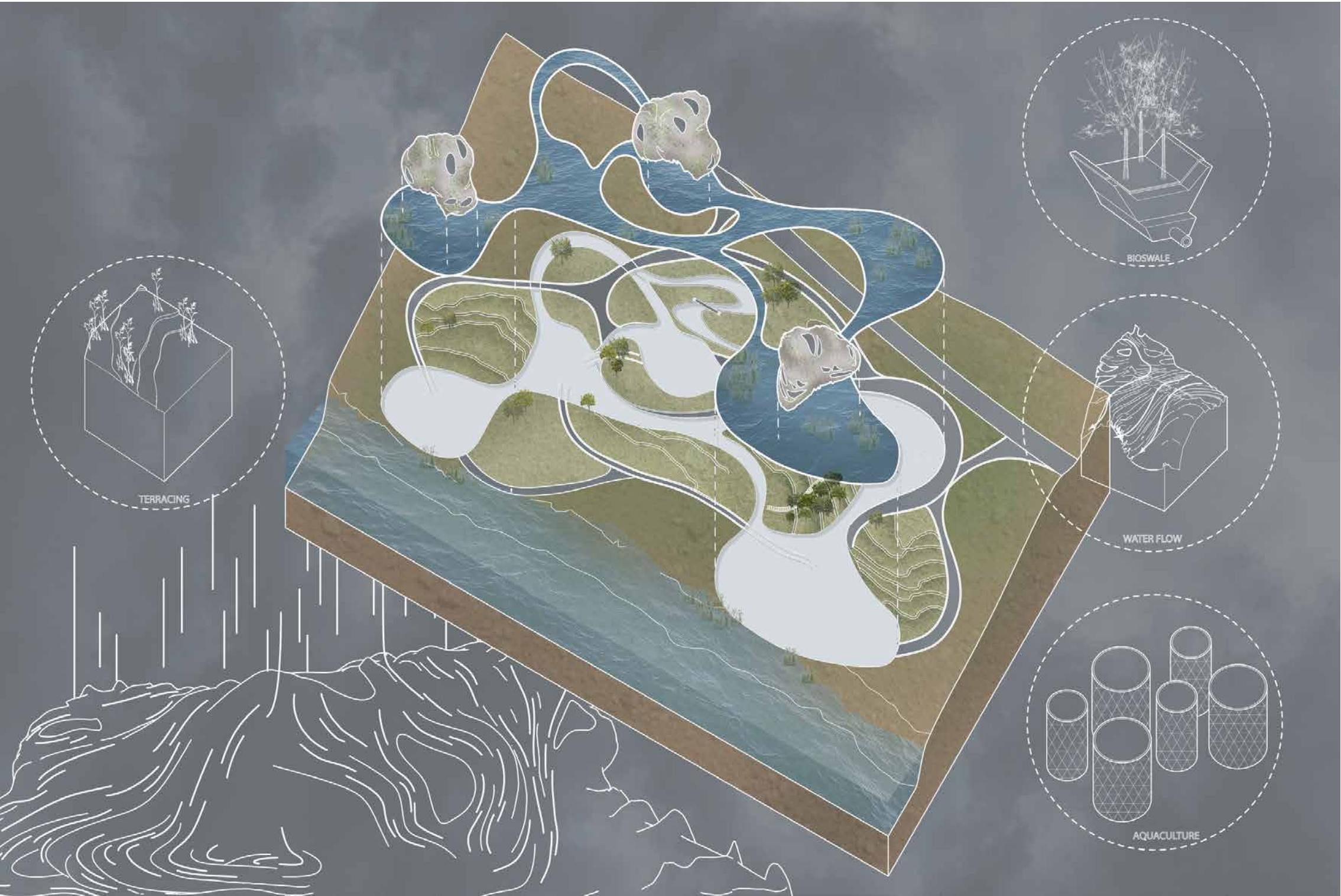




This project is a self sufficient halfway house designed to integrate the inhabitants with the landscape, and to create a community to empower the residents to become self sufficient, to provide for the themselves as well as the community.

Slow-Grow treats the architecture as a continuation of the landscape rather than as an intervention on it. The building and the landscape are designed in tandem with each other to perform as the eco machine in a self-sustained ecosystem that is able to support both itself and the people who are occupying the structure. A texture map of water flow lines is applied to the combined surface of the nearby landscape and building form. The ridges are used as guides for water channeling and window openings.



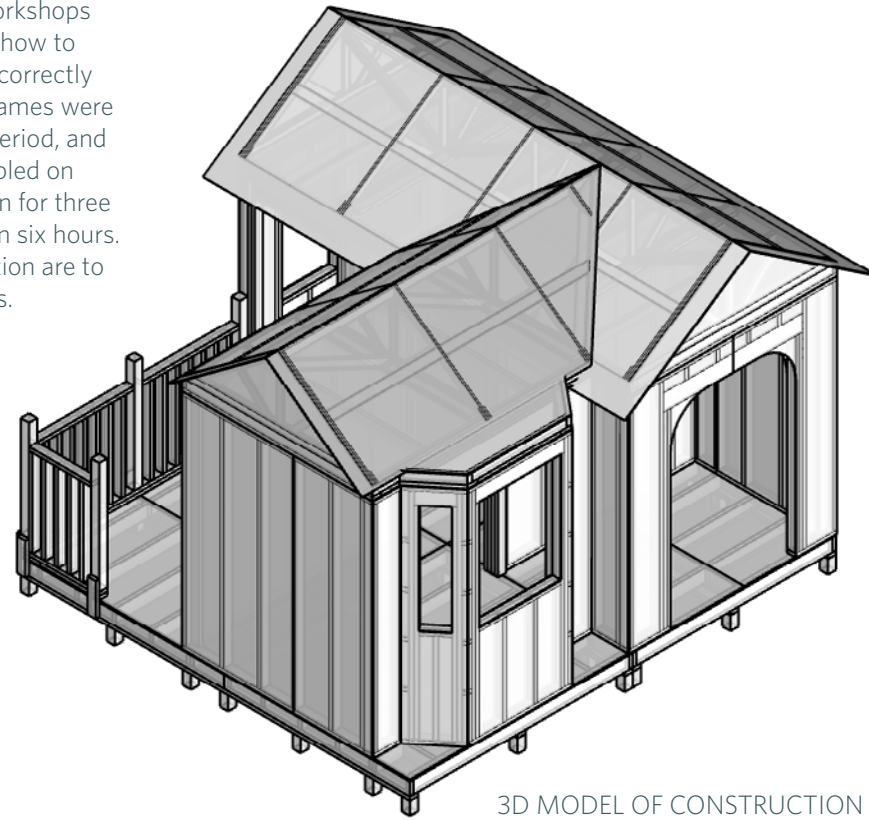




SCD Booth - Animal Crossing was a six-month long project to build a wood frame booth where event attendees could walk through to enjoy themed decorations, and play a collaborative game. Rather than having a simple box design, our team incorporated a bay window, curved steps, an open back porch, and a decorative pitched roof. These elements created a more unique spatial experience, and combined with the attentiveness to craftsmanship, really set our booth apart.

As the Architecture Chair, I was responsible for the formal design and structure -- how each and every piece of lumber comes together to ensure a safe and stable booth. Each frame was modeled exactly as it should be constructed, and made into drawing blueprints for members to follow.

I also led construction workshops where I taught members how to operate power tools and correctly build frames. All of the frames were built over a four-month period, and the structure was assembled on site in six days, to be open for three days, and disassembled in six hours. All lumber in good condition are to be used in following years.



First Place
Overall



Environmental Award Environmental Award
Energy Use On-Site Practices

